

Workflow Example

This document describes a workflow example for HTCondor and is intended to help cluster users getting started.

The goal is to submit a job with HTCondor, which runs a Docker container from a previously built image, executes a shell script to rename the GPUs, and then runs a short Python script, printing out the used CUDA device into a text file.

Note: make sure to replace all occurrences of <username> with your actual LST username.

Download the example project

Create a `test_gpu` directory on your machine and download the following files into it.

```
test_gpu.sub
test_gpu.sh.txt
test_gpu.py.txt
test_gpu_dockerfile.txt
```

Change into that directory from your terminal.

Note: Since the Wiki does not allow to upload certain file extensions, I had to rename the files in order to upload them.

Please change the names of the files by removing all `.txt` extensions and rename `test_gpu_dockerfile.txt` to just `dockerfile`. (Currently looking for a different solution)

Docker image

Follow the instructions here to create a new project in the LST registry, using your LST username as the name for the project.

You need to build a docker image from the provided dockerfile and push it to the LST registry.

The LST docker images must follow a certain naming convention: `docker.coli.uni-saarland.de/[project_name]/[image_name]:[tag]`

Here, we will use the following:

```
[project_name] : your LST username
[image_name]   : test_gpu
[tag]          : v1
```

Build the docker image (Docker needs to be installed on your machine):

```
docker build -t docker.coli.uni-saarland.de/<username>/test_gpu:v1 .
```



Login to the LST registry:

```
docker login docker.coli.uni-saarland.de
```



Push the image which you've built to the LST registry:

```
docker push docker.coli.uni-saarland.de/<username>/test_gpu:v1
```



Now, you can check the website interface of the registry. You should see the image in your project (username).

Bash script

When submitting a job, you need to provide a bash script which HTCondor will run inside your Docker container. A bash script for this example project is provided as `test_gpu.sh`. This script renames the GPUs, which is necessary for them to be accessible via `CUDA_VISIBLE_DEVICES`, and then executes the Python script below.

Python script

The Python script `test_gpu.py` checks if there is a GPU available and prints the output into a txt file. Since there is no output path specified, the txt file will appear under `/nethome/<username>`.

Submit file

The submit file specifies a number of things which HTCondor needs to know, such as the docker image to create a container from, the bash script which should be executed in the container, where to store log files, and the resource requirements of your job. The only thing you need to change, is to replace all occurrences of `<username>` with your LST username.

Tip: if you're using Vim, you can replace all occurrences at once by

```
:%s/exp1/exp2/g
```



where `exp1` is `<username>` and `exp2` your actual LST username.

Copy Files

From your terminal, copy the `test\_gpu` directory into your home directory on the server.

```
scp -r test_gpu <username>@login.coli.uni-saarland.de:~
```



Log in

From the university network, you can directly log into the submit server via

```
ssh <username>@submit.coli.uni-saarland.de
```



If you work remotely, you either need to log in to the login server first:

```
ssh <username>@login.coli.uni-saarland.de  
ssh submit
```



Or you can use a proxy jump.

Log File Paths

HTCondor will create log files at the paths specified in your submit file. The relevant lines are *output*, *error* and *log*. If the paths don't exist already, you need to create them:

```
mkdir -p /scratch/<username>/logs/test_gpu
```



Submit

Before you submit, make sure that the bash script `test_gpu.sh` is executable. Test:

```
ll /nethome/<username>/test_gpu/test_gpu.sh
```



Make it executable:

```
chmod u+x /nethome/<username>/test_gpu/test_gpu.sh
```



Now, you're ready to submit. From the submit server, change into the `test_gpu` directory. Then submit your job:

```
condor_submit test_gpu.sub
```



You can monitor your jobs by typing

```
condor_q
```



If your job is on hold, you can get more information by typing

```
condor_q -hold <jobid>
```



 user/cluster/c_example-workflow.txt  Last modified: 2025/06/24 14:51 by saguilar

LST-Wiki
